

Haoran Lei

Philadelphia, PA | +1 267-592-2887 | hr782357345@gmail.com

EDUCATION

University of Pennsylvania, MSE Electrical Engineering

Philadelphia, PA

- From Sep 2025 to Dec 2026 (Expected)
- GPA: 3.9/4.0
- Relevant Coursework: Smart Devices, Statistics for Data Science, Graph Neural Networks, Nanorobotics, Introduction to Robotics, Feedback Control Design and Analysis, Applied Machine Learning

University of Edinburgh, BEng (Hons) Electronics and Electrical Engineering, First Class

Edinburgh, UK

- From Sep 2021 to Jun 2025
- Final Project: Implementation of a Brain-Computer Interface Model on Raspberry Pi
- Relevant Coursework: Analog Circuits, Digital System Design, Microelectronic Design and Fabrication, Software and Embedded Systems, Signals and Communication Systems, Control and Instrumentation Engineering, Sensors and Instrumentation, Microelectronic Devices, Power Electronics and Electrical Machines

WORK EXPERIENCE

Tashan Technology

Beijing, China

Robotics Software Intern / Robotics & Embodied AI Intern

Jun 2026 – Aug 2026

- Integrated a RealSense D435i RGB-D camera, tactile sensors, robotic arms, end effectors, and an RK3568 edge-computing platform into a custom mobile dual-arm robotic system.
- Quantized perception models, converted them to RKNN format, and deployed them on the RK3568 platform for onboard inference.
- Configured the ROS Noetic and MoveIt stack, including robot models, joint limits, controllers, planning groups, and communication between the mobile base, dual arms, cameras, and end effectors.
- Diagnosed hardware-software integration issues involving sensor calibration, coordinate transformations, control interfaces, power-up behavior, and unstable robotic operation.

Lighttech Engineering SL

Elx, Spain

Summer Intern

Aug 2024 – Sep 2024

- Independently developed the complete hardware architecture of a wireless intelligent joystick for automotive and engineering-control applications, covering joystick position sensing, processing, display, wireless communication, power and charging, and USB Power Delivery interfaces.
- Designed schematics and PCB layouts, selected key components, and integrated deadman safety logic, processor interfaces, display circuitry, and communication modules within a constrained enclosure.
- Reviewed power budgets, interface constraints, and safety-trigger logic while evaluating signal integrity, manufacturability, and maintainability.
- Assembled and powered up the prototype, calibrated joystick zero points and travel ranges, and debugged power-supply, signal-acquisition, display, and communication issues.
- Collaborated with mechanical and electrical engineers to optimize component placement, enclosure space, and cable routing, advancing the design from concept to a functional prototype.

PROJECT EXPERIENCE

Inflatable Soft Wearable Assistive Device, University of Pennsylvania (GRASP Lab, Sung Robotics Lab)

Philadelphia, PA

Master's Thesis Project and Lead Student Researcher at Penn

Nov 2025 – Present

- Collaborate with Stanford University's CHARM Laboratory on the cross-institutional development of an inflatable soft robotic assistive device.
- Lead the hardware architecture, experimental design, and prototype development by integrating IMUs, pressure sensors, solenoid valves, pneumatic components, and control units.

- Develop an IMU-based sensing system to track body posture and motion and provide real-time input for assistance triggering and control decisions.
- Implement pressure acquisition and closed-loop feedback control to monitor chamber pressure and regulate pneumatic inflation.
- Design and integrate solenoid-valve driving and control logic that coordinates posture estimation, pressure feedback, airflow regulation, and pneumatic actuation.
- Conduct sensor calibration, valve-control integration, and stability, compliance, and repeatability testing while maintaining experimental records and parameter-tracking documentation.

Implementation of a Brain-Computer Interface Model on Raspberry Pi

Edinburgh, UK

Bachelor's Thesis Project

Sep 2024 – Jun 2025

- Developed a non-invasive brain-computer interface using EEG and EOG signals and implemented filtering, normalization, segmentation, model training, and embedded inference.
- Trained and evaluated RNN and LSTM models, achieving approximately 96% validation accuracy, and deployed real-time inference on a Raspberry Pi 5.
- Designed a GPIO switch and LED PCB that connected the embedded hardware with the inference and control pipeline while optimizing latency, memory usage, and thermal performance.

HypED – Student Technology Society Developing Hyperloop Pod

Edinburgh, UK

Electronics Navigation Project Manager

Sep 2023 – Aug 2024

- Led the integration of a Raspberry Pi 4 with optical-flow, full-spectrum, and acceleration sensors for pod-speed estimation and wireless control.
- Designed the navigation mainboard, including power regulation, Raspberry Pi connections, accelerometer circuitry, and multiple sensor interfaces.
- Completed circuit simulation, schematic design, and PCB layout for a time-of-flight distance-sensing system, optimizing sensor power, controller interfaces, signal stability, and maintainability.
- Performed PCB soldering, power-up inspection, sensor communication testing, and distance-measurement debugging, resolving hardware-connection, power-supply, and data abnormalities.
- Developed system-testing and troubleshooting procedures addressing sensor drift, power stability, and wireless-link reliability, and completed integrated vehicle validation.

IMU Glove-Controlled Smart Vehicle

Philadelphia, PA

Electronics Navigation Project Manager

Sep 2025 – Nov 2025

- Developed a dual-device architecture consisting of an ESP32-based wearable glove and an STM32-based vehicle controller, using an IMU to map hand orientation and motion to driving commands.
- Integrated motor control, lighting effects, and wireless command reception using the STM32 as the central vehicle controller.
- Integrated an ultrasonic sensor for forward obstacle detection and implemented basic autonomous obstacle avoidance while completing circuit connection, chassis assembly, sensor calibration, and system testing.

SKILLS

Programming and Embedded Development: C, C++, Python, Verilog, Bash, Linux, Git

Embedded Platforms: STM32, ESP32, Raspberry Pi 4/5, RK3568/RK3588, GPIO, embedded Linux, edge deployment, embedded debugging

Hardware Design: Schematic capture, PCB layout and routing, component selection, circuit simulation, soldering, board bring-up, sensor interfaces, power regulation, Altium Designer, LTspice, Fusion 360

Sensors and Robotics: IMUs, pressure sensors, optical-flow sensors, time-of-flight sensors, accelerometers, ultrasonic sensors, RealSense D435i, tactile sensors, ROS Noetic, MoveIt, feedback control, sensor integration, pneumatic actuation

Development Tools: Vivado, Xilinx SDK, RViz, Gazebo, ROS command-line tools

LANGUAGES: Mandarin Chinese (Native), English (Fluent), Spanish (B2)